Kuri Site Pipeline — Workflow by Stages

Updated Nov 3, 2026 • **Authoritative Operational Runbook** • End-to-End Governance from GIS File to Supplier Handoff

A construction site resides at exactly one stage at any time. The 6-digit Site ID (`CBE-XXXXXX`) remains immutable from creation through client delivery.

Stage 1: Upload (Admin Portal — Developer)

ADMIN PORTAL • /ADMIN

The GIS team uploads a 12-column file carrying no Site ID. Kuri mints an immutable 6-digit Site ID (`CBE-XXXXXX`).

- **Cadastral Auto-Resolution (TNGIS WFS):** Point-in-polygon ray-casting against GeoServer `cadastral_analysis:fmb_ulpin` auto-populates Survey Number, Sub-division Number, and ULPIN, canonicalizing Revenue Village and Taluk.
- **Derived Signals:** Google Maps link, total plot area converted to sq ft, 1,500 sq ft eligibility pre-check, and 40m duplicate scan.
- **Alternative Entry Points:** QA users creating manual records or field verifiers spotting new sites also mint 6-digit Site IDs and join the pipeline at QC.

Status on arrival: **QC Pending**

Stage 2: Desk QC (Data QC Portal — QC Analyst)

DATA QC PORTAL • /AUDITOR/DASHBOARD

The analyst reviews high-resolution satellite imagery (VHR / Cartosat) and Google Street View:

- **Evidence Capture:** Records observable construction signs (rebar, foundations, scaffolding), imagery source/date, and screenshots.
- **Plot Size Check:** Confirms total plot area against 1,500 sq ft threshold.
- **Builder Contacts:** Researches public signboards and registries; records contact with source URL.
- **Nearby Outcome:** If construction is visible near the pin but not at it, the analyst creates a separate child record with its own 6-digit Site ID linked through `Parent Site ID` .

Status on exit: **QC Done**

Stage 3: QA Review (Data QC Portal — QA Reviewer)

DATA QC PORTAL • /AUDITOR/DASHBOARD

A second reviewer independently audits the desk QC findings before field travel is permitted:

- **Three Distinct Outcomes:**
  - `Approved` : Releases the site for field assignment.
  - `Needs Correction` : Reopens Desk QC for imagery or coordinate re-examination.
  - `Disapproved` : Marks site `Not Eligible` (resting state; re-checked when fresh imagery arrives).

Status on approval: **Approved**

Stage 4: Assignment (Admin Portal — Operations)

ADMIN PORTAL • /ADMIN

Operations dispatches field verifiers. **Gated strictly on QA Approval**, preventing wasted travel.

- **Queue Priority:** Dispatched in order of Priority Tier (`Tier-1` , `Tier-2` , `Tier-3`).

- **Target Coordinates:** Verifiers are routed strictly to QC-confirmed coordinates, never raw detection points.

Status on assignment: **Field Pending**

Stage 5: Field Visit (Field Verification App — Field Verifier)

FIELD PORTAL • /APP/

Verifier travels to site. App activates GPS, logging captured coordinates, accuracy radius, and deviation from QC pin.

- **Conditional Requirements Logic:**
  - If *Construction Actively Happening* == '*Yes*': `Type of Construction` and `Stage of Construction` are **Mandatory**.
  - If *Construction Actively Happening* == '*No*': `Type` and `Stage` are **Optional/Disabled**; verifier selects structured *Inactivity Reason* (Vacant Land, Stalled, Demolished).
- **Structured Contacts:** Owner Contact and Contractor/Architect Contact are captured into separate slots with strict 10-digit mobile validation (`^[6-9]\d{9}$`).
- **Evidence:** Geotagged camera photos required. Verifiers can cross-verify TNGIS Cadastral survey markers.

Status on submission: **Completed** or **Closed**

Stage 6: Revisit and Reaudit Loops

FIELD PORTAL + DATA QC PORTAL

Ensures zero data loss while preserving immutable Site IDs:

- **Field Revisit:** Dispatched when details were incomplete, gate was locked, or telecaller flagged a *Wrong Number* (routing back to field for fresh phone acquisition).
- **Desk Reaudit:** Reopens QC when ground verifier reports coordinates discrepancy or mismatch.

Stage 7: Telecalling (Telecaller Portal — Telecaller)

TELECALLER PORTAL • /TELECALLER/DASHBOARD

Telecallers qualify leads, confirm construction specs, and gather commercial interest:

- **Queue Gate:** Only leads with at least one verified 10-digit mobile number appear in telecaller queue.
- **Standard Timezone:** All call logs and timestamps standardized in **Indian Standard Time (IST / Asia/Kolkata)**.
- **Structured Attempts:** Max 3 call attempts across morning, afternoon, and evening windows.
- **Dispositions:** `CONSTRUCTION_CONFIRMED` , `FOLLOW_UP` , `NOT_INTERESTED` , `WRONG_NUMBER` .

Stage 8: Package & Deliver (Admin Portal — Operations)

ADMIN PORTAL • /ADMIN/EXPORT

Generates high-value, executive client delivery packages (Excel reports):

- **Comprehensive Attributes:** 6-digit Site ID, Old Unique ID cross-reference, TNGIS Survey/Sub-div, verified address, stage, floors, contact numbers with stakeholder role designation, and data freshness date.
- **Suppression Controls:** Privacy filters withhold unverified contacts and declined leads.